Group 12

Evaluating Traditional and Large Language Model Approaches for Aspect-Based Sentiment Analysis of Business Reviews

Zack Mosley
Halıcıoğlu Data Science Institute
School of Computing, Information, and Data Sciences
University of California San Diego
La Jolla, CA, USA
ORCID

Jillian O'Neel
Halıcıoğlu Data Science Institute
School of Computing, Information, and Data Sciences
University of California San Diego
La Jolla, CA, USA
ORCID

***Abstract*—Customer reviews contain valuable insights into specific aspects of a business, but their large volume makes manual analysis difficult. In this work, we perform aspect-based sentiment analysis (ABSA) on Google Maps reviews using three approaches: TF-IDF with logistic regression, a fine-tuned RoBERTa transformer model, and a generative large language model (GPT-4.1-mini). We evaluate all models using both weakly supervised labels and a manually annotated gold-standard dataset.**

Keywords*—*aspect-based sentiment analysis, machine learning, TF-IDF, transformer models, RoBERTa, large language models, GPT, weakly supervised learning, natural language processing, customer review analysis

***GitHub*—https://github.com/zmosDS/google_review_insights**

I. Introduction

For businesses that provide products and/or services, customer reviews on platforms such as Google Maps are valuable sources of insight. These reviews capture authentic customer experiences at a scale that is difficult to replicate through traditional methods, such as surveys or focus groups. They often include actionable feedback on product quality, staff interactions, pricing, and ambience, all of which can inform business improvements. However, the scale of modern review platforms is both a strength and a challenge. Manually analyzing large volumes of customer feedback is labor-intensive, time-consuming, and prone to inconsistent interpretation. Data science techniques provide a way to automate this analysis, enabling businesses to extract insights consistently and prioritize improvements more effectively.

Sentiment analysis (SA), also known as opinion mining, is the computational study of opinions, attitudes, or emotions expressed toward an entity [1]. It is commonly framed as a classification task that determines the polarity of sentiment in written natural language (e.g., positive, negative, neutral). A more fine-grained extension of SA is aspect-based sentiment analysis (ABSA), which combines sentiment detection with aspect identification [2]. Rather than assigning a single sentiment label to an entire document, ABSA identifies specific topics and determines the sentiment expressed toward each topic individually. This more granular approach better reflects natural language, where speakers often express both positive and negative opinions within the same review, as illustrated in Fig. 1.

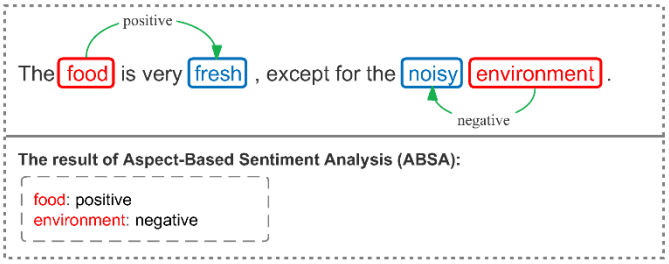

Fig. 1. Example of ABSA applied to a sentence. [2]

Because customer reviews frequently contain mixed feedback on multiple aspects of a business, assigning a single overall sentiment label is overly simplistic. In many cases, similar insight can already be inferred from the accompanying 1-5 star rating. ABSA provides a more informative approach by extracting specific aspects and their associated sentiments. Applying ABSA to a large collection of customer reviews enables businesses to systematically identify strengths and weaknesses in key areas such as product quality, service, and pricing, thereby supporting more targeted and data-driven decision-making.

II. Model Pipeline

In this work, the input to the analysis was the text of a customer review. Three different modelling approaches were applied to output aspect-sentiment pairs (e.g., *product quality:positive*, *service:negative*), as illustrated in Fig. 2. The models evaluated include (1) a classical machine learning baseline using TF-IDF with logistic regression, (2) a transformer-based neural encoder (RoBERTa), and (3) a generative large language model (ChatGPT-4.1-mini).

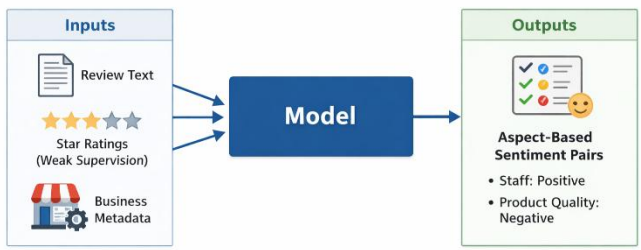

Fig. 2. Basic diagram of model inputs and outputs.

III. Related Work

Research in aspect-based sentiment analysis (ABSA) has evolved from rule-based natural language processing techniques to classical machine learning models and, more recently, to deep learning and large language model approaches.

Early work in ABSA relied on rule-based natural language processing techniques. Hu and Liu proposed one of the earliest feature-based opinion mining systems, which extracted product aspects using the frequency of noun phrases and determined sentiment polarity using a predefined opinion lexicon [3]. Similarly, Thet et al. extended this approach by incorporating dependency relationships to better associate opinion words with specific aspects in text and by cleverly introducing a domain-specific sentiment polarity lexicon [4]. A key strength of rule-based methods is their interpretability: aspect-sentiment pairs are derived from explicit rules and lexicons, making system decisions transparent. These approaches also require little or no labeled training data. However, rule-based methods rely heavily on explicit aspect mentions and struggle to identify implicit aspects, such as "It fits nicely in my pocket," which implicitly refers to product size [3]. In addition, lexicon-based approaches often fail to capture contextual nuances such as sarcasm and may not generalize well to informal language commonly found in online reviews. In this work, a rule-based method is used to rapidly label millions of reviews with aspect-sentiment pairs in order to generate weak supervision data for more advanced models.

Later approaches introduced supervised machine learning models such as support vector machines (SVMs) and logistic regression to learn aspect–sentiment patterns from labeled data. For example, in response to the SemEval-2014 Aspect-Based Sentiment Analysis task, Kiritchenko et al. used supervised models, primarily SVMs, trained on features such as n-grams and TF-IDF representations to identify aspect terms and classify sentiment expressed toward those aspects [5]. These approaches capture richer lexical patterns than rule-based systems and can be trained relatively quickly with modest computational resources. However, they rely on manually engineered features such as tokenization, n-gram selection, and text normalization, and they often struggle to capture deeper contextual relationships such as sarcasm. The TF-IDF with logistic regression baseline used in this paper is conceptually similar to the approach of Kiritchenko et al., as both treat ABSA as a supervised text classification task [5].

More recent work has applied deep learning models to ABSA. Wang et al. introduced attention-based neural networks that allow models to focus on words most relevant to a given aspect [6]. Later, Devlin et al. introduced BERT, a transformer-based language model pretrained on large corpora that learns contextualized bidirectional representations of text [7]. Models such as RoBERTa build upon this architecture and have become a common and high-performing approach for ABSA tasks [8]. Transformer models generate contextualized word embeddings, meaning that a word's representation depends on surrounding context, enabling the model to capture nuanced sentiment expressions and relationships between words [7]. These models also require little manual feature engineering because they learn representations directly from raw text. However, transformer models require significant computational resources and their predictions are often less interpretable than those of simpler models.

More recently, researchers have explored generative large language models for ABSA. Simmering and Huoviala evaluated models such as GPT-3.5 and GPT-4 on the SemEval-2014 ABSA task and found that prompt-based LLM approaches can achieve performance comparable to traditional models [9]. Like transformer-based classifiers, LLMs have strong language understanding and require minimal feature engineering. However, they are computationally expensive, slower to run than traditional models, and can be costly to apply at scale.

This progression highlights the evolution of ABSA methods toward increasingly context-aware models and motivates the comparison of classical machine learning, transformer-based models, and generative LLM approaches in this study.

IV. Dataset and Features

For this project, we used the Google Local Data (2021) dataset, compiled by researchers at the UCSD McAuley Lab, including Tianyang Zhang and Jiacheng Li [10, 11, 12]. This dataset consists of over 650 million Google Maps reviews and corresponding business metadata spanning all 50 U.S. states as of September 2021. Because the full dataset exceeds 200 GB, this study restricts the analysis to a single state, California, which still includes over 70 million reviews across approximately 500,000 businesses.


```
{
  'user_id': '106533466896145407182',
  'name': 'Amy VG',
  'time': 1568748357166,
  'rating': 5,
  'text': "I can't say I've ever been excited about a dentist visit
before, but there's a first for everything! Loved my experience at
Lush today. Every person in the office was friendly and personable-
plus the office itself is gorgeous! Great experience, I highly
recommend!",
  'pics': [
  {
    'url':
['https://lh5.googleusercontent.com/p/AF1QipMBzN4BJV9YCObcw_ifNzFPm-
u38hO3oimOA8Fb=w150-h150-k-no-p']
  },
  {
    'url':
['https://lh5.googleusercontent.com/p/AF1QipNS1PEXEvadfUlhRkRDJ09id
    Mxh3CveZGZYuTo5=w150-h150-k-no-p']
  }
  ],
  'resp': {
    'time': 1568770503975,
    'text': 'We love getting to meet new patients like yourself.
Thanks for giving our office a chance to take care of your dental
needs and thanks for the nice review!'
  },
  'gmap_id': '0x87ec2394c2cd9d2d:0xd1119cfbee0da6f3'
}
```


Fig. 3. An example review entry in JSON format [10]

```
{
  'name': 'Walgreens Pharmacy',
  'address': 'Walgreens Pharmacy, 124 E North St, Kendallville, IN
46755',
  'gmap_id': '0x881614ce7c13acbb:0x5c7b18bbf6ec4f7e',
  'description': 'Department of the Walgreens chain providing
prescription medications & other health-related items.',
  'latitude': 41.451859999999996,
  'longitude': -85.2666757,
  'category': ['Pharmacy'],
  'avg_rating': 4.2,
  'num_of_reviews': 5,
  'price': '$$',
  'hours': [['Thursday', '8AM-1:30PM'], ['Friday', '8AM-1:30PM'],
['Saturday', '9AM-1:30PM'], ['Sunday', '10AM-1:30PM'], ['Monday',
'8AM-1:30PM'], ['Tuesday', '8AM-1:30PM'], ['Wednesday', '8AM-
1:30PM']],
  'MISC': {
    'Service options': ['Curbside pickup', 'Drive-through', 'In-store
pickup', 'In-store shopping'],
    'Health & safety': ['Mask required', 'Staff wear masks', 'Staff
get temperature checks'],
    'Accessibility': ['Wheelchair accessible entrance', 'Wheelchair
accessible parking lot'],
    'Planning': ['Quick visit'],
    'Payments': ['Checks', 'Debit cards']
  },
  'state': 'Closes soon · 1:30PM · Reopens 2PM',
  'relative_results': ['0x881614cd49e4fa33:0x2d507c24ff4f1c74',
'0x8816145bf5141c89:0x535c1d605109f94b',
'0x881614cda24cc591:0xca426e3a9b826432',
'0x88162894d98b91ef:0xd139b34de70d3e03',
'0x881615400b5e57f9:0xc56d17dbe420a67f'],
  'url':
'https://www.google.com/maps/place//data=!4m2!3m1!1s0x881614ce7c13acb
  b:0x5c7b18bbf6ec4f7e?authuser=-1&hl=en&gl=us'
}
```


Fig. 4. An example business metadate entry in JSON format [10]

The dataset is organized by state and contains two JSON files per state: one containing customer reviews (*reviews.json*) and one containing business metadata *(metadata.json*). The *reviews.json* file includes fields such as *user id*, *time*, *rating*, *text*, and *gmap id*, where *text* contains the review content, *rating* is a 1-5 star score, and *gmap id* is a Google-provided unique identifier for each business (Fig. 3). The *metadata.json* file contains business information such as *gmap id*, *name*, and *description* (Fig 4).

The data cleaning pipeline in this work constructs a combined California reviews dataset by merging review and business metadata files while retaining only columns relevant for downstream analysis. Reviews are preserved using a left join on the shared business identifier (*gmap_id*). To manage memory constraints, the merged dataset was written to disk in fixed-sized JSON chunks. Duplicate reviews, reviews with missing ratings, non-English reviews and reviews with empty or missing text were removed during preprocessing.

For modeling, the dataset was further refined by selecting the relevant features *user id*, *rating*, *text*, and *gmap id*. Observations were also filtered based on business and user activity levels. Specifically, only users with 5-250 total reviews and businesses with at least 10 reviews were retained to reduce noise from sparse users and potential spam activity while ensuring sufficient review signal for each business. A unique *review_id* was generated for each review using a SHA-256 hash of concatenated metadata fields to maintain traceability throughout downstream text processing. Exploratory data analysis revealed that the rating distribution is strongly skewed toward positive reviews, with more than 80% of reviews having ratings of four stars or higher (Fig 5). This class imbalance was considered during model training.

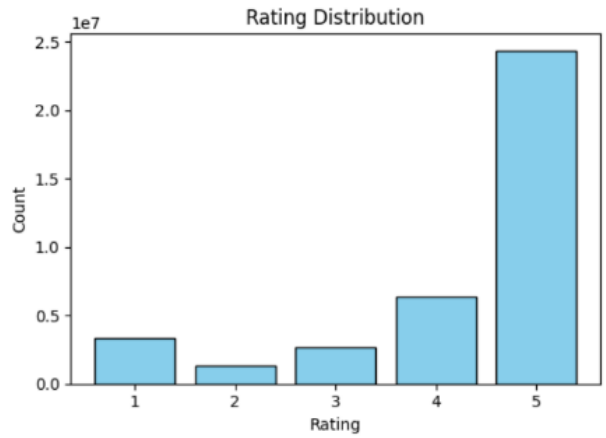

Fig 5. Count of reviews per rating, indicating skew towards positive reviews.

The dataset does not include aspect-based sentiment labels. Therefore, a rule-based algorithm was used to rapidly generate aspect-sentiment labels to support weak supervision during model training. Word frequency analysis was used to identify six common business aspects and a corresponding dictionary of related key words was constructed (Fig 6.) A *general* aspect category was also created to capture cases where sentiment was expressed without reference to a specific aspect (e.g., "This place is great!"). The aspect dictionary was designed such that all aspects appeared across both positive and negative reviews (Fig. 7). Sentiment polarity was assigned using a simple threshold based on the review rating in which aspects extracted from reviews with ratings greater than three were labeled as positive, while aspects from reviews with ratings less than three were labeled as negative.

```
# Define aspects and keywords
aspect_keywords = {
    'product_quality': ['food', 'meal', 'dish', 'taste', 'flavor', 'delicious', 'fresh',
                        'menu', 'order', 'coffee', 'drink', 'burger', 'chicken', 'pizza',
                        'product', 'quality', 'item', 'purchase', 'bought', 'broken'],
    'service': ['service', 'staff', 'employee', 'manager', 'worker', 'associate',
                'waiter', 'waitress', 'server', 'cashier', 'friendly', 'rude',
                'helpful', 'unprofessional', 'told', 'said', 'asked'],
    'wait_time': ['wait', 'waiting', 'slow', 'fast', 'quick', 'minutes', 'hour',
                  'line', 'queue', 'busy', 'long', 'forever', 'delay', 'prompt'],
    'price_value': ['price', 'prices', 'expensive', 'cheap', 'cost', 'value', 'worth',
                    'affordable', 'overpriced', 'reasonable', 'pricey', 'money', 'deal'],
    'cleanliness': ['clean', 'dirty', 'filthy', 'hygiene', 'gross', 'spotless',
                    'mess', 'messy', 'tidy', 'maintained'],
    'atmosphere': ['atmosphere', 'ambiance', 'decor', 'seating', 'crowded', 'quiet',
                   'noisy', 'comfortable', 'cozy', 'parking', 'location', 'space'],
    'general': ['great', 'good', 'bad', 'worst', 'best', 'amazing', 'awesome',
                'excellent', 'horrible', 'terrible', 'awful', 'love', 'hate',
                'disappointing', 'perfect', 'recommend', 'experience']
}
```

Fig. 6. Aspect key word dictionary

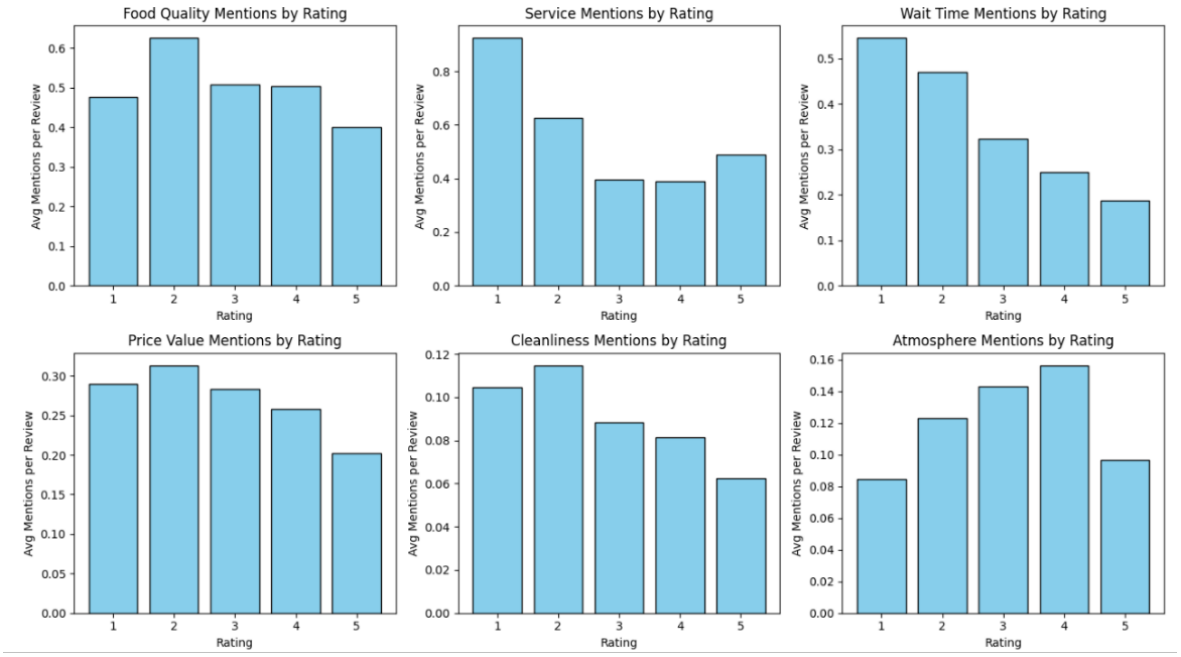

Fig. 7. Aspect mentions by rating.

This rule-based labelling process enabled rapid annotation of approximately 23 million reviews for use in model training. However, this approach does not capture nuanced cases where a single review contains mixed feedback across multiple aspects. To address this limitation, a manually annotated gold-standard dataset was created for model validation. Reviews were first tokenized into sentences and a subset of reviews was sampled with balanced representation across rating levels. A custom annotation interface was developed to support efficient human labeling. The interface presents reviews one sentence at a time and allows annotators to assign aspect-sentiment pairs using predefined controls (Fig 8). Sentence-level annotations were subsequently aggregated to produce review-level ABSA labels paired with the original review text.

Fig. 8. Screenshot of annotation interface.

Although manual annotation better captures mixed feedback and complex language patterns such as sarcasm, aspect-level sentiment analysis remains inherently subjective. In many ABSA studies, multiple annotators label the same data and disagreements are resolved through discussion or majority voting [13]. Due to resource constraints, this multi-annotator approach was not implemented in this study. Instead, annotation guidelines were followed during labeling and more than 2,500 reviews were painstakingly labeled to construct the evaluation dataset. Model performance is therefore evaluated on both the weakly labeled dataset and the manually annotated gold-standard dataset.

Several predictive tasks could be performed using the Google Reviews dataset, including predicting the star ratings from review text, detecting potentially fraudulent or spam reviews, or classifying business types. However, this study focuses on aspect-based sentiment analysis because it provides more granular insights by identifying which specific elements of a business drive positive or negative customer feedback. Additionally, ABSA presents a more technically interesting challenge due to the linguistic complexity of reviews.

V. Methods

This section describes the learning algorithms used to perform aspect-based sentiment analysis on the review dataset. Three modeling approaches of increasing complexity were evaluated: a classical machine learning baseline using TF-IDF features and logistic regression, a transformer-based neural network (RoBERTa), and a generative large language model (GPT-4.1-mini). All modeling and data processing were implemented in Python using the scikit-learn [14], Hugging Face Transformers [15], pandas [16], and NumPy [17] libraries, with large language model inference conducted through the OpenAI API [18].

A. TF-IDF + Logistic Regression

We first implemented a classical machine learning baseline using TF-IDF features and logistic regression. This model provides a simple and computationally efficient reference point against which the more complex neural and generative approaches can be compared.

TF-IDF transforms raw review text into sparse numerical vectors by weighting each term by how frequently it appears in a document relative to how rarely it appears across the entire dataset [19]. Formally,

$$\mathrm{TFIDF}(t,d) = \mathrm{TF}(t,d)\log\left(\frac{N}{DF(t)}\right)$$

Terms that are locally prominent but globally distinctive receive the highest weights. The vocabulary was extended to include bigrams to preserve negation phrases such as "not good", which are critical for sentiment discrimination. Frequency thresholds were also applied to suppress rare noise terms and near-universal filler words.

Because each review may express sentiment across multiple aspect categories simultaneously, prediction was framed as a multi-label binary classification problem. Independent logistic regression classifiers were trained for each label. Each classifier estimates

$$P(y_k = 1 \mid x) = \sigma(w_k^T x + b_k)$$

and minimizes regularized cross-entropy loss [20]. The regularization strength λ was selected using cross-validation. Logistic regression is well-suited for this task because its sparse weight vectors directly identify the most predictive TF-IDF features for each aspect, while its probabilistic outputs allow threshold tuning to address the severe class imbalance present in aspect-level sentiment labels.

B. RoBERTa Fine-Tuning

Unlike TF-IDF representations which discard word order, RoBERTa (Robustly Optimized BERT Pretraining Approach) is a transformer-based encoder pretrained on large text corpora using masked language modeling [8]. During pretraining, the model learns to predict randomly masked tokens from surrounding context, enabling it to develop rich contextual word representations, where the same word carries different embeddings depending on the words around it. This directly addresses the contextual ambiguity that limits bag-of-words models. Input text was tokenized using byte-pair encoding (BPE), which segments words into sub word units that are mapped to learned embedding vectors [21]. Review sequences were truncated to a maximum length of 128 tokens.

The model was fine-tuned for multi-label aspect-based sentiment classification by adding a linear classification head on top of the final [CLS] token representation. This produces 14 independent sigmoid outputs, one for each aspect-sentiment label (Fig. 9). The model minimized binary cross-entropy loss across all labels jointly:

$$L = -\frac{1}{K}\sum_{k=1}^{K}[y_k \log \widehat{p_k} + (1-y_k)\log(1-\widehat{p_k})]$$

where K=14 is the number of labels.

Training was performed for three epochs using the AdamW optimizer with a learning rate of 2×10−5, a standard choice for transformer fine-tuning that preserves pretrained contextual representations while adapting to the target task [22]. A batch size of 64 to maximize GPU memory utilization.

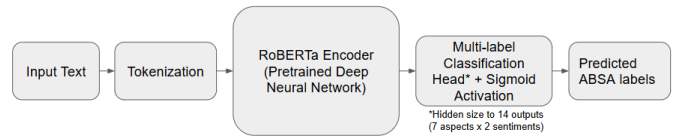

Fig. 9. Diagram of RoBERTa modeling pipeline.

C. Generative Large Language Model

While the previous approaches treat aspect-based sentiment analysis as a supervised classification problem, we also evaluated a generative large language model (LLM) that extracts structured information directly from the review text. Specifically, we used the GPT-4.1-mini model through the OpenAI API [18].

GPT-4.1-mini was selected because it provides strong performance on structured extraction tasks, reliable instruction following, and significantly lower cost and latency than larger models, making it practical for processing thousands of reviews via API calls [23]. Instead of predicting labels through a trained classifier, the model was prompted to read a review and generate structured aspect-sentiment pairs present in the text.

Given an input review, x, the model generates a sequence of output tokens

$$y = (y^1, y^2, \dots, y_T)$$

representing predicted aspects and their associated sentiment polarity. Like other transformer-based language models, GPT-4.1-mini generates tokens autoregressively by estimating the probability of each token conditioned on the input and previously generated tokens [19]:

$$P(y|x) = \prod_{t=1}^{T} P(y_t|x, y_{<t})$$

In practice, the model was prompted to return predictions in a structured JSON format, enabling aspect identification and sentiment classification to be performed jointly without additional task-specific training (Fig. 10).

```
DEVELOPER_PROMPT = """
You are an expert annotator for aspect-based sentiment analysis (ABSA).

Task:
Read one customer review and extract all aspect-sentiment pairs present.

Allowed aspects:
- Product Quality
- Service
- Wait Time
- Price/Value
- Cleanliness
- Atmosphere
- General

Allowed sentiment labels:
- positive
- negative
- neutral
- mixed

Rules:
1. Return only the requested JSON schema.
2. Include one entry per distinct aspect mentioned in the review.
3. Do not invent aspects not supported by the review.
4. Use General if there is an overall sentiment that does not clearly fit another aspect.
5. Keep evidence short and directly grounded in the review text.
6. If an aspect contains both positive and negative sentiment, use mixed.
7. If there is no clear aspect sentiment, return an empty list for pairs.
"""
```

Fig. 10. Example LLM prompt.

We evaluated the LLM using both zero-shot and few-shot prompting strategies. In the zero-shot setting, the model receives only a task description and the review text and must infer the desired output format without labeled examples. In the few-shot setting, the prompt additionally includes several example reviews with their corresponding aspect-sentiment annotations to guide the model's predictions. This prompting paradigm leverages the ability of large language models to perform new tasks with minimal supervision [25].

VI. Results & Discussion

A. Evaluation Metrics

Our primary evaluation metric is weighted F1, which balances precision and recall while accounting for class imbalance [19]. For a given label k, precision and recall are defined as:

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}$$

$$\text{Recall}_k = \frac{TP_k}{TP_k + FN_k}$$

where TP_k, FP_k, and FN_k represent the true positives, false positives, and false negatives for label k, respectively.

The F1 score for each label is defined as the harmonic mean of precision and recall:

$$F1_k = \frac{2 \cdot \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}$$

Because aspect-based sentiment analysis is a multi-label task with imbalanced label frequencies, we compute both macro F1 and weighted F1 across all labels. Macro F1 treats each label equally and is defined as:

$$F1_{macro} = \frac{1}{K}\sum_{k=1}^{K} F1_k,$$

where K is the number of labels.

Weighted F1 accounts for label imbalance by weighting each label's F1 score according to its number of true instances:

$$F1_{weighted} = \frac{\sum_{k=1}^{K} n_k F1_k}{\sum_{k=1}^{K} n_k},$$

where n_k is the number of true samples for label k and $N = \sum_{k=1}^{K} n_k$ is the total number of samples across all labels.

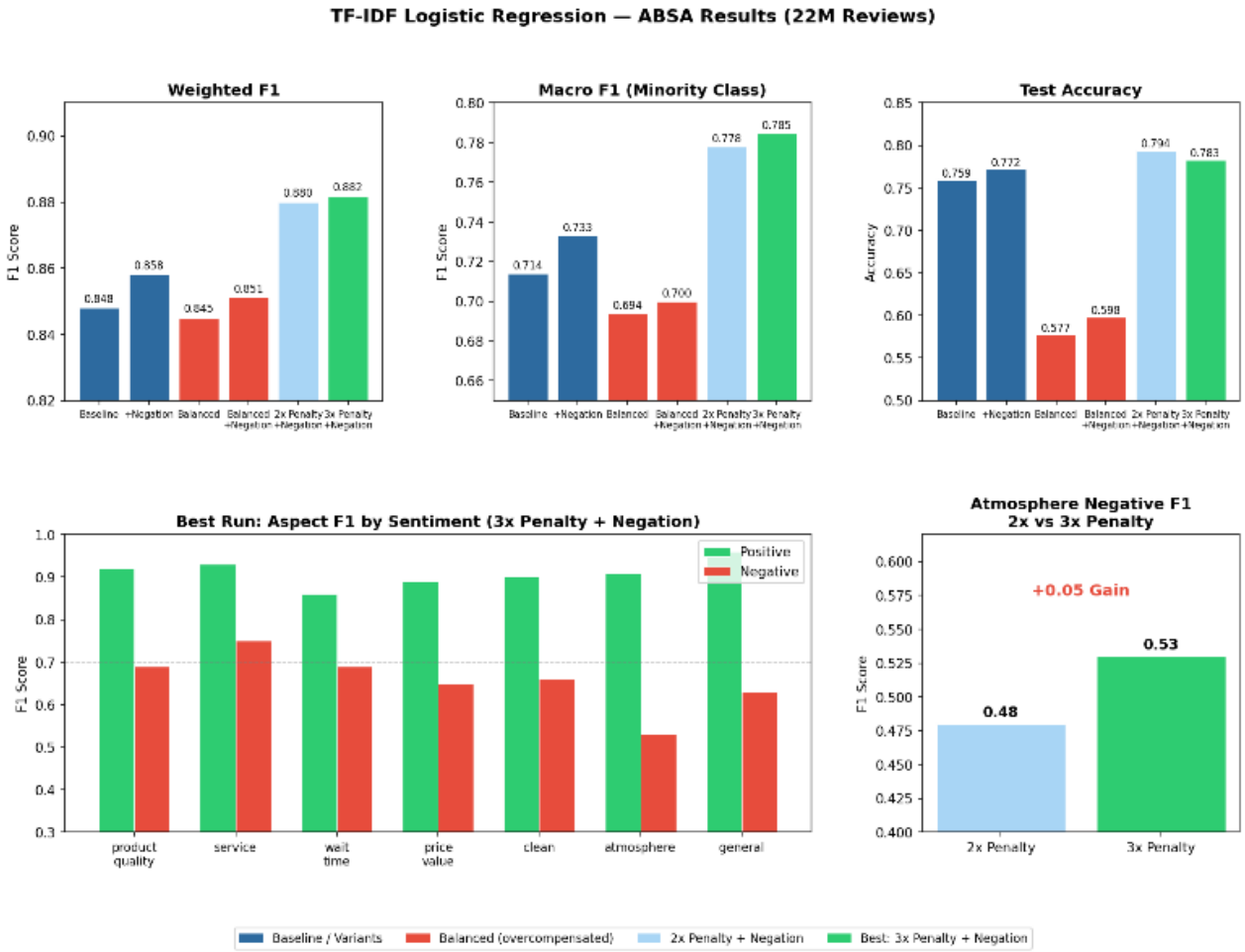

Fig. 11. ABSA results for TF-IDF + Logistic Regression

In our dataset, negative sentiment labels comprise approximately 18% of all label instances, meaning accuracy alone can be misleading. For example, a model that predicts only positive sentiment would achieve artificially high accuracy while failing to detect any negative cases.

For this reason, we report weighted F1, macro F1, precision, and recall for all three models to enable direct comparison. In addition, we analyze performance across individual aspect-sentiment categories to identify which labels each model predicts well and where errors occur.

B. TF-IDF + Logistic Regression

Review text was vectorized using a vocabulary of 10,000 features with unigram and bigram ranges. The vocabulary size was capped at 10,000 to balance coverage against dimensionality. A minimum document frequency of 500 was used to eliminate rare typos and domain-specific noise, while a maximum document frequency of 50% suppressed uninformative terms.

A MultiOutputClassifier was used to train 14 independent logistic regression models in parallel, one per aspect-sentiment label. This approach allows each classifier to learn label-specific decision boundaries. The regularization strength λ was selected using 3-fold cross-validation optimizing weighted F1, with maximum number of iterations set to 500 to ensure convergence.

Six configurations were evaluated on the weakly labeled dataset, progressively addressing class imbalance: a baseline model, baseline with negation tokens, balanced class weighting, balanced weighting with negation, and asymmetric penalty weighting of 2x and 3x for the minority class (both with negation). Balanced weighting overcorrected, collapsing precision to 0.23 while driving recall to 0.97. The 3x penalty with negation configuration produced the best performance, achieving macro F1 = 0.785, weighted F1 = 0.882, and test accuracy = 0.783.

Confusion matrices in Fig. 12 illustrate the variation in performance across labels. For example, *general:positive* achieved 61% recall, whereas *atmosphere:negative* reached only 17% on the validation set.

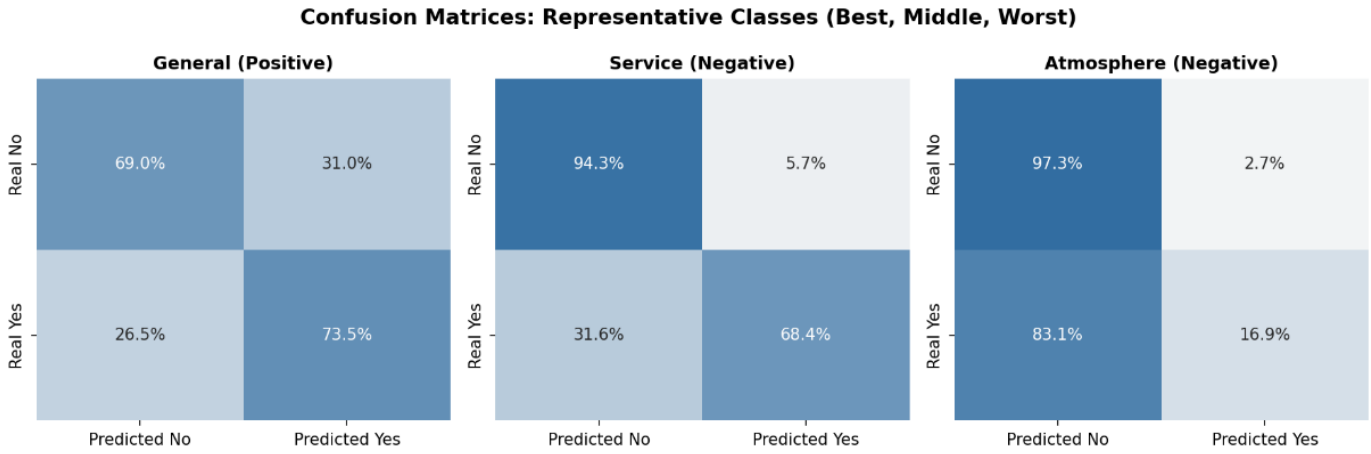

Fig. 12 Confusion Matrices for Representative Classes

The top predictive features for *atmosphere:negative* include ambiguous terms like "quiet" and "cozy" which appear in both sentiment classes. This highlights a limitation of bag-of-words models, which lack the contextual reasoning needed to resolve sentiment based on surrounding context.

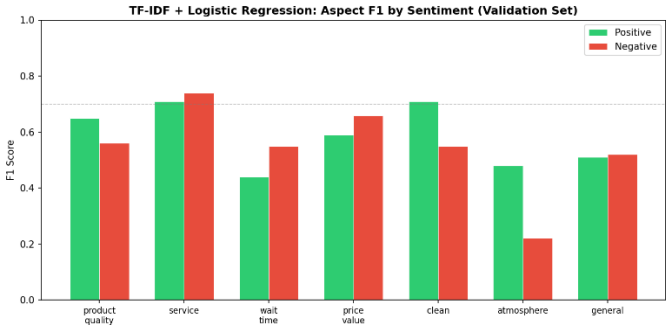

Fig. 13. TF-IDF + Logistic Regression, F1 by Aspect-Sentiment

When evaluated on the manually labeled gold-standard validation set, macro F1 decreased to 0.564, reflecting the gap between weakly-supervised training labels and true human annotations.

C. RoBERTa Fine-Tuning

RoBERTa was fine-tuned on the weakly labeled dataset using five training set sizes: 10k, 25k, 50k, 100k, and 250k samples. A learning rate of 2×10^{-5}, batch size of 64, and three training epochs were used across all runs. Because fine-tuning transformer models is computationally expensive, cross-validation was not performed. Instead, sample size served as the experimental variable across a fixed 90/10 train-test split.

Performance improved substantially as the training dataset increased in size. The 10k model achieved a macro F1 of only 0.359, with several labels scoring zero due to insufficient minority-class examples. By 100k samples, macro F1 reached 0.876 and weighted F1 reached 0.928. Performance gains plateaued beyond this point with the 250k model improving

macro F1 by only 0.008 while nearly tripling the training time. Based on this trade-off, the 100k model was selected as the final configuration.

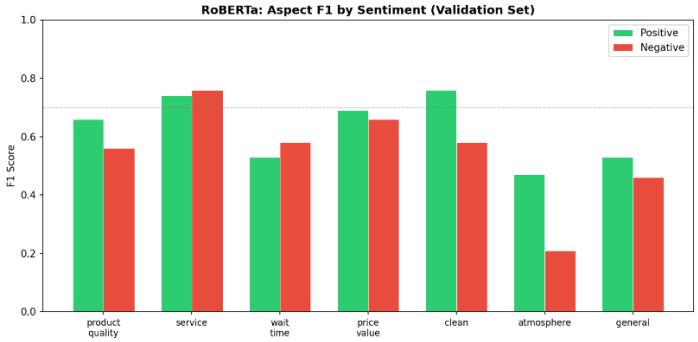

Fig. 14. RoBERTa, F1 by Aspect-Sentiment

Performance improvements were particularly notable for negative sentiment labels. All negative labels achieved F1 scores above 0.75 on the test set. In comparison, the TF-IDF model achieved only 0.53 F1 for *atmosphere:negative.*

Confusion matrices in Fig. 15 show recall for *atmosphere:negative* improving from 54.6% with TF-IDF to 70.6% with RoBERTa. This improvement reflects the ability of contextual embeddings to resolve sentiment ambiguity that bag-of-words features cannot capture.

However, when evaluated on the manually labeled gold-standard validation set, performance decreased to macro F1 = 0.584 and weighted F1 = 0.605. This result is consistent with the label noise observed in TF-IDF and suggests that both models partially learn patterns from weak supervision that do not fully transfer to human annotations.

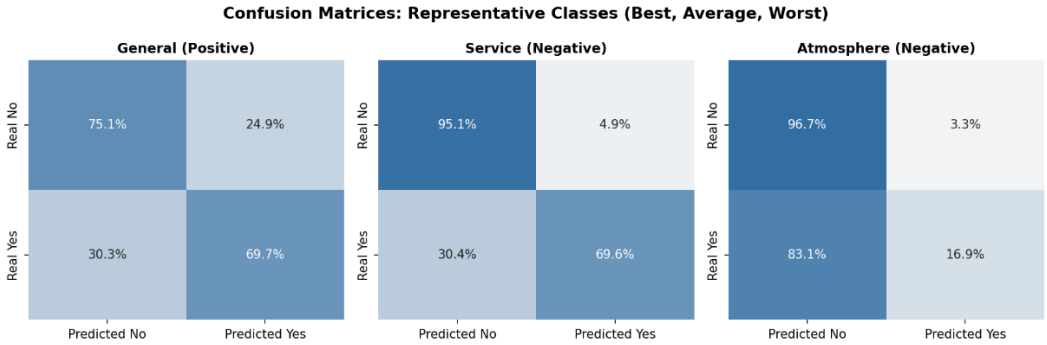

Fig 15. Confusion matrices for representative classes.

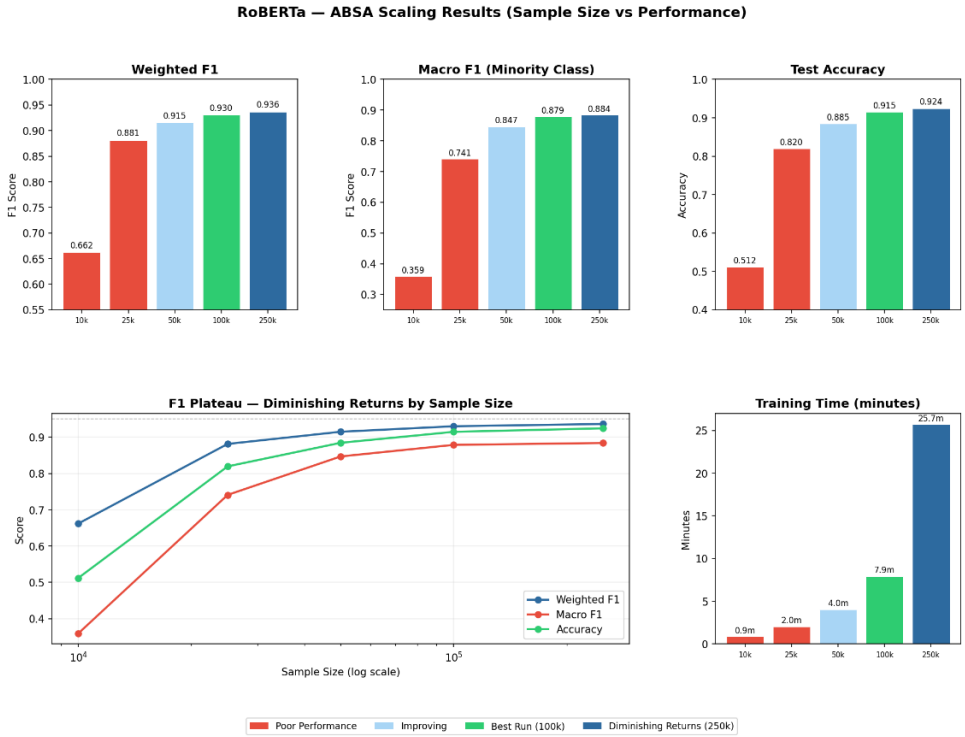

Figure 16. RoBERTa ABSA Scaling Results

D. Generative Large Language Model

A generative large language model approach was evaluated using GPT-4.1-mini via the OpenAI API. Two prompting strategies were tested: zero-shot prompting, which includes only task instructions and input review text, and few-shot prompting, which additionally includes seven labeled examples. Deterministic decoding (temperature = 0) was used to improve reproducibility. No task-specific cross-validation or finetuning was performed.

On the weakly labeled dataset, zero-shot and few-shot prompting achieved similar performance, with macro F1 scores of 0.518 and 0.515 respectively. Few-shot prompting increased precision but reduced recall, resulting in no meaningful improvement in overall F1. Performance was strongest for clear categories such as service sentiment while remaining weaker on subjective labels such as atmosphere

On the manually labeled gold-standard dataset, performance improved substantially. The zero-shot configuration achieved macro F1 = 0.749 and weighted F1 = 0.757. This suggests that the LLM aligns more closely with human interpretations of aspect–sentiment relationships than with the rule-based weak labels. The weak labels were generated using simple keyword rules tied to overall star ratings, whereas the LLM infers aspect–sentiment relationships directly from the semantic content of the review text. As a result, the model likely captures nuanced or mixed sentiments that disagree with the weak labels but align more closely with human judgement.

Cost and runtime considerations are also important. On the manually labeled review set, the zero-shot configuration consumed approximately 1.16 million tokens and cost around $0.75 to run via the OpenAI API, whereas the few-shot configuration consumed approximately 2.45 million tokens and cost about $1.25. Combined with stronger zero-shot performance, this indicates that few-shot prompting did not justify the additional token usage. Additionally, running both configurations on the 2,500 manually labelled reviews took approximately 75-80 minutes.

E. Model Comparison

Tables 1 and 2 summarize the performance of all three models on the validation dataset. RoBERTa achieved the highest performance on the weakly labeled dataset, while the generative LLM demonstrated stronger alignment with human annotations on the manually labeled evaluation set. The TF–IDF baseline, while computationally efficient, struggled to capture contextual sentiment cues and showed lower performance across most aspect categories.

Table I: Model Comparison on Manually Labeled Validation Set

Model	Weighted F1	Macro F1
TF-IDF + LogReg	0.595	0.564
RoBERTa (100k)	0.605	0.584
GPT-4.1 Zero-Shot	**0.757**	**0.749**

Table II: Per-Aspect F1 on Manually Labeled Validation Set

Aspect	TF-IDF Pos	TF-IDF Neg	RoBERTa Pos	RoBERTa Neg	GPT-4.1 Zero-Shot Pos	GPT-4.1 Zero-Shot Neg
Product Quality	0.65	0.56	0.66	0.56	**0.82**	**0.77**
Service	0.71	0.74	0.74	0.76	**0.91**	**0.87**
Wait Time	0.44	0.55	0.53	0.58	**0.62**	**0.72**
Price Value	0.59	0.66	0.69	0.66	**0.85**	**0.82**
Cleanliness	0.71	0.55	0.76	0.58	**0.91**	**0.77**
Atmosphere	0.48	0.22	0.47	0.21	**0.70**	**0.60**
General	0.51	0.52	0.53	0.46	**0.60**	**0.53**

VII. Conclusion

This work explored three approaches to aspect-based sentiment analysis (ABSA) on large-scale Google Maps review data: a classical TF-IDF + logistic regression baseline, a fine-tuned RoBERTa transformer model, and a generative large language model prompted via the OpenAI API. Among the models evaluated, the transformer-based RoBERTa model achieved the strongest performance on the weakly supervised training data, demonstrating the benefits of contextual language representations compared to simple word-frequency features. The TF-IDF baseline performed competitively but struggled to resolve context-dependent sentiment expressions. In contrast, the generative LLM achieved the strongest performance on the gold-standard manually labeled dataset, suggesting that its pretrained language knowledge aligns more closely with human annotations than with the rule-based weak labels.

Because RoBERTa was trained only on weakly labeled data in this study, its performance on the gold-standard dataset may have improved if the model had also been fine-tuned on the manually annotated reviews. Given the substantially higher computational cost and inference time of the generative LLM, even modest improvements in RoBERTa performance could make it the more practical approach for large-scale deployment.

Overall, this project demonstrated that machine learning techniques can effectively extract actionable insights from large scale customer review text. We successfully developed a complete analysis pipeline that includes cleaning and preprocessing, aspect selection, generation of both weakly labeled and manually labeled datasets, and evaluation of ABSA performance across models of increasing complexity. The results show that all three approaches were capable of identifying meaningful aspect-sentiment pairs that could be aggregated into useful metrics for business owners. Among the models tested, the fine-tuned RoBERTa transformer provides the most promising balance of predictive performance and computations efficiency.

Future work could explore fine-tuning the transformer models using larger manually labeled datasets, improving weak labeling strategies, or integrating the model into a business-owner dashboard that analyzes incoming reviews in near real-time to surface actionable insights about the customer experience.

VIII. Contributions

Zack Mosley led the data preparation and classical and neural modeling components of the project. His contributions included dataset selection, large-scale data cleaning, exploratory data analysis (EDA), and feature engineering for the ABSA pipeline. He developed and evaluated the TF-IDF + logistic regression baseline as well as the fine-tuned RoBERTa transformer model. Zack also contributed to project code organization and documentation, GitHub repository management including the README and report writing.

Jillian O'Neel led the research framing and generative modeling components of the project. Her contributions included conducting the literature review, designing the manual data labeling workflow and labeling widget, and performing manual annotation of review data. She implemented and evaluated the generative LLM approach for aspect-based sentiment analysis using the OpenAI API. Jillian also contributed to feature engineering, milestone deliverables, and report writing.

IX. Reflections

Although the results demonstrate that multiple machine learning approaches can extract meaningful aspect–sentiment patterns from large-scale review text, several areas could be improved if this project were repeated. One key limitation of this work was the relatively small size of the manually labeled dataset compared to the weakly supervised training data. While the gold-standard annotations provided valuable evaluation benchmarks, the dataset was too small to support effective supervised training for complex models such as RoBERTa. In future work, allocating additional time and resources to manual annotation would likely improve both model training and evaluation by providing higher-quality labeled examples. Expanding the gold-standard dataset would also enable more robust comparison between models trained on weak versus fully supervised data.

Another area for improvement involves the modeling approaches explored. For the generative LLM approach, we evaluated only a limited set of prompts and a single model configuration. With more time, we would experiment with additional prompt engineering strategies and alternative generative models to better understand their impact on aspect extraction performance. A particularly promising direction would be a hybrid workflow in which a generative LLM assists with labeling large volumes of reviews that are then verified or corrected by human annotators. This approach could significantly accelerate the creation of high-quality labeled datasets.

X. Response to Peer reviews

We thank both review groups for their thoughtful feedback and constructive suggestions. Below we summarize how we addressed the key points raised in the reviews.

First, both review groups noted that the generative LLM evaluation was incomplete in the milestone submission. In the final version of the project, we fully implemented and evaluated the generative LLM model using both zero-shot and few-shot prompting. The model was evaluated on both the weakly labeled dataset and the gold standard manually labeled dataset. We now report precision, recall, macro F1, and weighted F1 for both prompting strategies and directly compare these results with the TF-IDF and RoBERTa models.

Second, reviewers highlighted potential label noise introduced by using star ratings as weak sentiment labels. We acknowledge that assigning sentiment from review-level ratings can introduce noise, particularly for mixed or nuanced reviews. To address this limitation, we incorporated a manually labeled gold standard dataset and used it to evaluate all models in the final analysis. This helped quantify the gap between weakly supervised and human-labeled performance.

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